

blazar The most variable type of active galaxy, which includes BL Lacertae objects and the most violently variable quasars. See also *active galaxy*, *BL Lacertae object*, *quasar*.

blue shift The displacement of spectral lines to shorter wavelengths that occurs when a light source is approaching an observer. See also *Doppler effect*, *red shift*, *spectral line*.

Bok globule A compact dark nebula, roughly spherical in shape, which contains 1 to 1,000 solar masses of gas and dust and has a diameter of between 0.1 and a few light-years. Globules of this kind are believed to be cool concentrations of gas and dust that eventually will collapse to form protostars. They are named after Dutch-born astronomer Bart Bok, who made a detailed study of these objects. See also *protostar*.

brown dwarf star A body that forms out of a contracting cloud of gas in the same way as a star, but, because it contains too little mass, never becomes hot enough to ignite the nuclear-fusion reactions that power a normal star. With less than 8 percent of the Sun's mass, a brown dwarf glows dimly at infrared wavelengths, fading gradually as it cools down.

C

caldera A bowl-shaped depression caused by the collapse of a volcanic structure into an emptied magma chamber. A caldera is usually found at the summit of shield volcanoes such as those found on Venus, Earth, and Mars.

captured rotation See *synchronous rotation*.

carbonaceous chondrite see *chondrite*.

cataclysmic variable see *variable star*.

catadioptric telescope A type of telescope that combines mirror and lens components, rather than one or the other, to bring light to a focus. Schmidt-Cassegrain telescopes are a popular type of catadioptric telescope. See also *reflecting telescope*, *Schmidt-Cassegrain telescope*.

celestial equator A great circle on the celestial sphere that is a projection of Earth's own equator onto the celestial sphere. See also *celestial sphere*, *great circle*.

celestial poles The two points at which the line of Earth's axis, extended outward, meets the celestial sphere and around which the stars appear to revolve. The north celestial pole lies directly above Earth's North Pole and the south celestial pole directly above Earth's South Pole. See also *celestial sphere*.

celestial sphere An imaginary sphere that surrounds Earth. As Earth rotates from west to east, the sphere appears to rotate from east to west. In order to define the positions of stars and other celestial bodies, it is convenient to think of them as being attached to

the inside surface of this sphere. See also *celestial equator*, *celestial poles*.

center of mass The point within an isolated system of bodies around which those bodies revolve. Where the system consists of two bodies (for example, a binary star), it is located at a point on a line joining their centers. If both bodies have the same mass, the center of mass lies midway between them, whereas if one body is more massive than the other, it lies closer to the more massive of the two.

Cepheid variable A type of variable star that increases and decreases in brightness in a regular, periodic way. Cepheids are pulsating variables, which vary in brightness as they expand and contract. The more luminous the Cepheid, the longer its period of variation. See also *variable star*.

Chandrasekhar limit The maximum possible mass for a white dwarf star. If the mass of a white dwarf exceeds this limit, which is about 1.4 solar masses, gravity will overwhelm its internal pressure and it will collapse. The limit was first calculated by Indian astrophysicist Subrahmanyan Chandrasekhar in 1931. See also *white dwarf star*.

charge-coupled device (CCD) An electronic imaging device that consists of a large array of tiny light-sensitive elements. The image of an object is constructed by reading off the electrical charges that accumulate in each element during an exposure.

chondrite A stony meteorite that contains a large number of small, spherical objects called **chondrules**. A **carbonaceous chondrite** is one that is rich in carbon, carbon compounds, and volatile materials. Carbonaceous chondrites are thought to be some of the least-altered primitive remnants of the protoplanetary disk from which the solar system formed. See also *meteorite*, *protoplanetary disk*.

chromosphere The thin layer in the Sun's atmosphere that lies between the photosphere (the visible surface) and the corona. Its faint, reddish-pink light can be seen directly during a total eclipse of the Sun when the Moon hides the dazzling photosphere. See also *photosphere*.

circumpolar A term used to describe a star, or other celestial body, that remains above the horizon at all times when viewed from a particular place on Earth's surface.

circumstellar disk A flattened, disk-shaped cloud of gas and dust that surrounds a star. A disk of this kind is usually associated with a young or newly forming star, in which case it is composed of material from the original dusty gas cloud that collapsed to form the central star. See also *protoplanetary disk*.

closed universe A universe that is curved in such a way that space is finite but has no discernible boundary (analogous to the surface of a sphere).

A universe will be closed if its average density exceeds a particular value called the **critical density**. In the absence of a repulsive force, a closed universe will eventually cease to expand and will then collapse. See also *flat universe*, *open universe*, *oscillating universe*.

coma The cloud of gas and dust that surrounds the nucleus of a comet and which comprises its glowing "head." See also *comet*.

comet A small body composed mainly of dust-laden ice that revolves around the Sun, usually in a highly elongated orbit. Each time it approaches the Sun, gas and dust evaporate from its nucleus (the solid core of the comet) to form an extensive cloud, called the coma, and one or more tails. See also *coma*, *tail*.

conjunction A close alignment in the sky of two celestial bodies, which occurs when both bodies lie in the same direction as viewed from Earth. When a planet lies directly on the opposite side of the Sun from Earth, it is said to be at **superior conjunction**. If a planet passes between Earth and the Sun (Mercury and Venus are the only planets that can do this), it is said to be at **inferior conjunction**. See also *opposition*.

constellation One of 88 regions of the celestial sphere. Each constellation contains a grouping of stars joined by imaginary lines to represent a figure. The constellations are officially referred to by the Latin names of these figures. Many have been named after mythological characters or creatures (such as Orion, the Hunter) but some after more mundane objects (for example, Sextans, the Sextant). See also *asterism*.

continuous spectrum see *spectrum*.

convection The transport of heat by rising bubbles or plumes of hot liquid or gas. In a **convection cell**, rising streams of hot material cool, spread out, and then sink down to be reheated, so maintaining a continuous circulation.

core (1) The dense central region of a planet. (2) The central region of a star within which energy is generated by means of nuclear-fusion reactions. (3) A dense concentration of material within a gas cloud.

Coriolis effect The tendency of a wind or current to be deflected from its initial direction as a consequence of a planet's rotation. In the case of Earth, the deflection is to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

corona The outermost region of the atmosphere of the Sun or a star. The solar corona has an extremely low density and a very high temperature (about 2–9 million degrees Fahrenheit/1–5 million degrees Celsius). It cannot be observed except during a total eclipse of the Sun. See also *eclipse*, *solar wind*.

coronal mass ejection A huge, rapidly expanding bubble of plasma that is ejected from the Sun's corona. Containing billions of tons of material in the form of ions and electrons, together with associated magnetic fields, a typical coronal mass ejection propagates outward through interplanetary space at a speed of several hundred miles (kilometers) per second. See also *corona*, *ion*, *plasma*.

cosmic microwave background radiation (CMBR) Remnant radiation from the Big Bang, which is detectable as a faint distribution of microwave radiation across the whole sky. See also *Big Bang*.

cosmic rays Highly energetic subatomic particles, such as electrons, protons, and atomic nuclei, that hurtle through space at speeds close to the speed of light.

cosmological constant An extra term in Einstein's relativity equations which, if it has a positive value, corresponds to a repulsive force that could cause the universe to expand at an accelerating rate. Modern cosmologists associate the constant with a quantity called **vacuum energy** (residual energy that, according to quantum theory, exists even in a vacuum), one of the possible forms of the dark energy believed to permeate the universe. See also *dark energy*.

cosmological red shift see *red shift*.

cosmology The study of the nature, structure, origin, and evolution of the universe.

crater A bowl- or saucer-shaped depression in the surface of a planet or satellite, or at the summit of a volcano. Many have raised walls and some have a central peak. An **impact crater** is one excavated by an meteorite, asteroid, or comet impact, whereas a **volcanic crater** is the cavity from which a volcano discharges material. Raised walls are created by accumulation of ejected material.

critical density see *flat universe*.

crust The thin, rocky, outermost layer of a planet or major planetary satellite, which, like Earth, has separated into several layers, with the densest material toward its center and the least dense at its surface.

D

dark energy A little-understood form of energy that appears to comprise about 70 percent of the total amount of mass and energy in the universe. It exerts a repulsive effect and is believed to be causing the expansion of the universe to accelerate. See also *accelerating universe*.